Research on Desertification Monitoring and Vegetation Refinement Extraction Methods Based on the Synergy of Multisource Remote Sensing Imagery

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***Abstract*— Due to over-exploitation by humans and global climate change, desertification has become an increasingly severe issue, seriously threatening the stability of ecosystems and the sustainable development of resources. Therefore, this study focuses on the Hangjin Banner region in Inner Mongolia, using satellite remote sensing and remote aerial vehicles (RAV) remote sensing technology. Through wide-area coverage, long-term monitoring, multiscale analysis, and high-precision interpretation, the study demonstrates the strong synergistic effects of "multiscale interpretation" and "data fusion applications," systematically carrying out desertification monitoring grading and refined vegetation extraction. First, to address the problem that the information dimension of a single index is insufficient and it is difficult to reflect the development trend of desertification, the normalized difference vegetation index (NDVI)-albedo feature space applicable to the desert environment is inversely performed based on Landsat 8 satellite images from 2009 to 2023. Then, on the basis of the feature space, the desertification difference index (DDI), which realizes the wide-area desertification monitoring grading and spatio-temporal evolution analysis of the study area, and the hue-saturation-lightness greenway enhanced vegetation index (HSLGEVI), which has stronger applicability and stability in desert environments, were constructed based on the HSL color space and the hue tuning algorithm. This index can effectively overcome the limitations of the RGB vegetation index, clearly delineate the canopy edge of desert vegetation, and accurately extract surface meadow vegetation with lower chlorophyll content. To test the effectiveness of the HSLGEVI, the widely used and validated excess green index (EXG), vegetation difference vegetation index (VDVI), modified green-red vegetation index (MGRVI), and red–green–blue vegetation index (RGBVI) were selected for comparison. The results show that the accuracy of HSLGEVI is better than that of other indices, with overall accuracy and $F1$-score remaining above 90%. It reduces the impact of the RGB color space vegetation index on the accuracy of vegetation extraction, effectively overcoming misclassification and omission issues, and providing a reliable monitoring mechanism for desertification control in the Hangjin Banner area.**

***Index Terms*— Desertification difference index (DDI), hue-saturation-lightness greenway enhanced vegetation index (HSLGEVI), multisource remote sensing collaborative monitoring, normalized difference vegetation index (NDVI)-albedo feature space, vegetation refinement extraction.**

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I. Introduction

LAND desertification refers to the gradual degradation of the productivity and ecological functions of soil systems under the combined effects of natural and anthropogenic factors [1], [2]. This phenomenon usually stems from the interaction between irrational economic activities and fragile ecosystems, leading to the destruction of soil texture, structure, and biodiversity, which triggers large-scale land degradation and ultimately presents a desert-like surface pattern [3], [4], [5]. Land desertification disasters not only severely constrain local economic development but also exert great pressure on grassland ecosystems, leading to a serious lack of water resources and substantial degradation of ecosystem services, which in turn pose a serious threat to agricultural productivity and socioeconomic stability [6], [7], [8]. Therefore, establishing a dynamic desertification monitoring mechanism to effectively respond to the continuous challenges of desertification disasters, realize real-time monitoring of land degradation processes, and comprehensively assess the effectiveness of

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governance, are crucial [9], [10]. By recognizing the early signs of desertification in time and taking appropriate interventions, the degradation trend can be effectively slowed, ecosystem functions can be maintained, and the sustainable development of ecosystems can be promoted.

In recent years, scholars worldwide have conducted extensive research on land desertification and have committed to establishing high-precision and high-stability remote sensing monitoring systems. In the field of satellite remote sensing, Symeonakis and Drak [11] developed a desertification monitoring system that systematically monitored desertification in sub-Saharan Africa from four perspectives: normalized difference vegetation index (NDVI), rainfall utilization efficiency (RUE), surface runoff, and soil erosion. The development of the system has improved the accuracy of desertification monitoring and provided a scientific basis for environmental management in the areas concerned. Lv et al. [12] conducted long-term quantitative monitoring of desertification in the Qinghai Province based on long-time-series MODIS data products by inverting the enhanced vegetation index (EVI) as a key indicator of desertification change, which provides an analytical basis for desertification research in the region. Serrano [13] explored the ability of the NDVI vegetation index to assess ecological quality indicators in different regions and further elaborated on the prospects of the application of this vegetation index for future vegetation cover monitoring. Hua and Fan [14] verified the ability of the NSVI index and NDVI index to estimate the vegetation cover, respectively, and the results showed that the applicability and validity of the NDVI index were significantly better than that of the NSVI index in the areas with lower vegetation cover. Amir et al. [15] assessed surface reflectance through surface feature attributes, revealing the contributions of different surface features to surface reflectance and providing important technical support for environmental monitoring and management. Bright et al. [16] assessed the cover of forested areas at high altitudes using the albedo index, which provided insights into environmental changes in high-altitude forested areas and an effective method for monitoring changes in alpine forest ecosystems. Zeng et al. [17] used the NDVI index for vegetation biomass, topsoil grain size index (TGSI) for the landscape pattern of the study area, and surface albedo for micrometeorological conditions, respectively, and their results showed that the linear correlation results of TGSI index and albedo index can be used to represent the land desertification process in the study area. Wang [18] assessed land degradation conditions by analyzing the relationship between different degrees of desertification and topsoil texture to construct a TGSI. In the field of remote aerial vehicles (RAV) vegetation monitoring, Evstatiev et al. [19] performed an accurate assessment of aboveground biomass by improving the modified green–red vegetation index (MGRVI) and verified its validity for biomass estimation. Zhang [20] constructed the normalized green red vegetation index (NGRVI). This index can not only explore the influence of different plant characteristics on the vegetation extraction ability, but also verify the effectiveness of vegetation characteristics in constructing the corresponding vegetation extraction method. Hu and Li [21] extracted vegetation information of arable land through the excess green index (EXG) vegetation index, showing its potential application to vegetation index in different environments. Morlin et al. [22] combined machine learning algorithms to extract vegetation information and predict cover using the red–green–blue vegetation index (RGBVI) vegetation index, thereby demonstrating the potential application of visible light vegetation index technology in vegetation identification and coverage prediction and, in turn, the prospect of wide application of visible vegetation index technology in vegetation identification and coverage prediction. These studies have laid a solid foundation for land desertification monitoring and vegetation refinement extraction, further promoting the application and development of ecological monitoring techniques. In addition, in the field of ecological monitoring by satellite remote sensing, most of the studies rely on a single indicator such as to assess the spatial and temporal changes of desertification. However, due to the insufficiency of information dimensions, these single indicators have greater limitations in assessing desertification, and it is difficult to comprehensively reflect the multidimensional characteristics and development trends of desertification [23], [24]. Desertification is a complex process involving multiple aspects such as climate change, land degradation, water resource changes, and biodiversity [25]. A single indicator usually can only reflect one of these dimensions, and it is difficult to comprehensively present the dynamic changes and regional differences of desertification [26]. In addition, factors such as seasonal variations, environmental factors, and spatial heterogeneity can also affect the interpretation of indicators, resulting in an inaccurate assessment of the desertification process [27]. The evolution of desertification is also affected by the complex interactions between natural factors and anthropogenic activities, and a single index cannot effectively integrate these factors, thus limiting the accuracy and comprehensiveness of desertification monitoring [4], [28]. Therefore, it is difficult to achieve effective results in monitoring land desertification by relying only on a single index method.

For RAV visible light vegetation extraction, existing studies usually only utilize vegetation indices in RGB color space [29]. However, under the complex environmental conditions of deserts, the application of vegetation indices consisting of only RGB color channels will face greater limitations, mainly due to the high degree of correlation and interdependence among the R, G, and B color channels in the RGB color space, which makes it difficult to carry out quantitative analysis [30], [31], [32]. In particular, under extreme lighting conditions, this correlation is further exacerbated, resulting in problems such as overexposure or loss of detail in highlights, deviation of detail because of high contrast between strong light and shadows, and color distortion because of light reflection and high light intensity. This leads to a significant reduction in the vegetation extraction capacity of these vegetation indices in desert environments. In addition, in the field of remote sensing synergistic applications, a team of researchers from the University of Rennes, France, defined the integration of remote sensing technologies as four synergistic strategies [33]. These include “data comparison,” “multiscale interpretation,” “model calibration data comparison,”

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"multiscale interpretation," "model calibration," and "data fusion applications." Among these, "data comparison" shows weak synergy, while the other three categories show strong synergy. The above studies usually use only a single remote sensing monitoring tool, such as high-resolution satellites or low-altitude visible RAVs, to monitor spatial and temporal changes in desertification. Although some studies have attempted to combine satellite and RAV data, such combination mostly remains at the stage of "data comparison" with weak synergy, failing to effectively explore and utilize the respective advantages of the two technologies and resulting in relatively limited synergistic effects. In summary, single-technology or shallow fusion methods have failed to give full play to the wide-area, long-time series, multiscale, and high-precision interpretation potential of multisource remote sensing technology for desertification monitoring.

This study leverages the synergistic effects of "multiscale interpretation" and "data fusion application" to comprehensively utilize high-resolution satellite remote sensing and high-precision visible light RAV imagery for desertification classification and vegetation fine extraction in the study area. First, high-resolution satellite imagery is used to construct a desertification difference index (DDI) to grade the degree of desertification and identify key desertified areas. Subsequently, based on high-precision RAV visible light remote sensing imagery, a hue-saturation-brightness greenway EVI (HSLGEVI) is constructed using tone adjustment algorithms and channel mixing enhancement techniques, enabling the fine extraction of regional vegetation information and the assessment of the land desertification process. This collaborative monitoring approach fully utilizes the advantages of satellite remote sensing, including broad-area coverage and multitemporal monitoring, as well as the centimeter-level high precision of RAV imagery. It allows for rapid identification of key desertification areas in the study region and facilitates high-precision monitoring and vegetation extraction. Furthermore, the HSLGEVI vegetation index demonstrates better applicability in desert environments, accurately capturing vegetation features even under strong lighting conditions. This index model effectively avoids the loss of high-brightness details and color distortion under extreme lighting conditions, reducing misjudgments and omissions in vegetation indices based on the RGB color space in complex lighting environments.

II. Materials and Methods

A. Study Area

1) Overview of the Study Area: Hangjin Banner, located in the central part of the Inner Mongolia Autonomous Region, is a typical arid and semi-arid area situated in one of the desertification hotspots in northwestern China, and is a typical land degradation-affected region. Due to its fragile ecological environment, scarce water resources, and harsh climatic conditions, the region has long been severely impacted by wind and sand erosion as well as soil erosion, leading to significant land degradation. In addition, with the intensification of human activities, Hangjin Banner is facing increasingly severe desertification issues, manifested in the degradation of land resources,

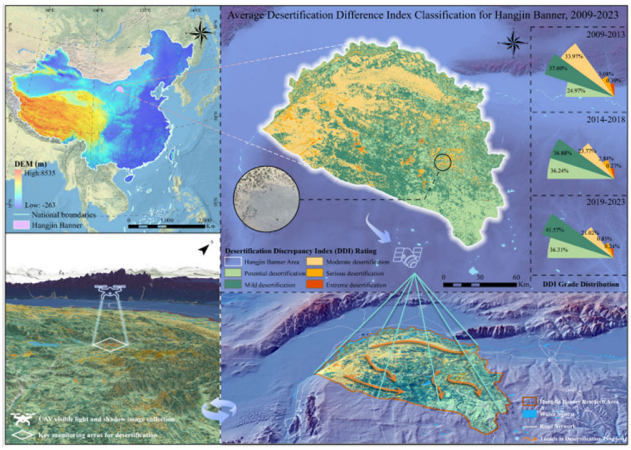


Fig. 1. Trends in land desertification development in the study area of Hangjin Banner.

a decline in vegetation cover, worsening soil erosion, and frequent sandstorms, all of which severely threaten the stability and sustainability of the local ecosystem [34], [35], [36]. Therefore, the desertification problems in Hangjin Banner reflect the general characteristics of land degradation in the northwest region, making it highly typical, representative, and valuable for research. Moreover, Hangjin Banner is a key area for desertification control and ecological restoration projects in Inner Mongolia. Its management experiences provide valuable lessons for desertification control in other parts of northwestern China and have significant social value and demonstrative potential.

2) Geographic Attributes of the Study Area: Hangjin Banner lies at the junction of the Ordos Plateau and the Hetao Plain, characterized by a landscape of grasslands, sand dunes, and mountains. Its geographic coordinates range from 39°36′ N to 40°26′ N and 106°35′ E to 107°58′ E [37], [38]. The total area of the region is approximately 11 400 km^2, with an altitude ranging from 1300 to 1700 m above sea level. At the same time, the region has a temperate continental climate and is situated in the transition zone between arid and semi-arid climates [39], [40]. Precipitation is mainly concentrated in summer, whereas winter is relatively dry, with an average annual precipitation range of 144.6–350 mm. The study area of the Hanging Jade Banner is shown in Fig. 1.

Fig. 1 illustrates the average changes in soil erosion and its distribution by severity in the Hangjin Banner study area from 2009 to 2023. It also presents a 3-D model based on the elevation data of the study area, showing the trend of desertification progression. In addition, the figure includes the region where drone-collected samples were taken.

B. Research Data

1) Data Sources and Initial Preprocessing: This study was based on Landsat 8 OLI/TIRS satellite imagery with less than 3% cloud cover. The path/row numbers of the imagery used were 129/32, 128/32, and 128/33 (since the Landsat series satellite equipped with the TM sensor was launched on March 1, 1984, and decommissioned on June 5, 2013, while the Landsat series satellite equipped with the OLI sensor

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TABLE I
LIST OF RESEARCH DATA

Data type	Data attribute	Data sources
Landsat 8 OLI_TIRS. Landsat TM Satellite Imagery	Path/line:129/32;128/32; 128/33; The cloud cover is less than 5%	U.S. Geological Survey (USGS) (https://earthexplorer.usgs.gov/))
DEM data	-	ASTGTM (https://lpdaac.usgs.gov/products/astgtmv003/)
RAV visible light imagery	Spectral widths: Red : 650 ± 16 nm; Green: 560 ± 16 nm; Blue: 450 ± 16 nm	DJI Elf 4rtk Field collection in the study area

began its official operation and data service on May 30, 2013, the images from these two sensors were selected to ensure the accuracy and continuity of the data). The time series of the imagery covers the period from 2009 to 2023, with a major concentration in the months of June to September to rule out the influence of seasonal variability and differences in vegetation growth on the results, thus ensuring validity. The image data from the RAV were collected by employing a DJI Innovations (DJI) Wizard 4 RTK drone, flying at an altitude of around 200 m and a speed of 8 m/s, while utilizing a timed shooting mode. The overlap rates for the images were configured at 70% and 30%. Within a sample area of 1 × 1 km, roughly 120–130 images were taken when the drone flew at an altitude of 200 m, whereas about 50–80 images were recorded at an altitude of 300 m. Table I lists the names of the data used, the sources of the data, and the detailed attributes of the data.

Before data preprocessing, in order to ensure the consistency of the image data and the smooth progress of subsequent processing, the resolution of the remote sensing imagery is first standardized to 30 m, and the coordinate system is unified to WGS1984, enabling localized UTM projection processing. Then, to address potential small voids and occasional outliers in the mosaicked data, raster range detection is used to identify missing pixels, and attribute tables are employed to check for anomalous pixels. These missing and anomalous values can lead to inaccurate total area calculations when performing multiyear area statistics or creating transition matrices. To resolve this, a neighborhood statistical method is applied, where the mode of pixels within the neighborhood is used to fill in the missing and anomalous pixels, thereby correcting the area statistics bias. Finally, due to the misalignment of original pixels across multiple years, when clipping data from different years, errors may occur at the boundary pixels, which affects the accuracy of statistical results. Pixel snapping and alignment processing are applied to eliminate these errors and ensure the accuracy of area statistics.

2) Research Data and Preprocessing Methodology Flow: The data preprocessing process is as follows: first, radiometric calibration was applied to the images using the radiometric correction tool in ENVI 5.3.1. Wavelength-specific calibration was conducted, with BLL chosen as the output type and data formatted in floating-point precision to ensure calibration accuracy. Next, the fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASHs) algorithm was applied to correct for atmospheric scattering and absorption, thereby enhancing the spectral accuracy of the data. After atmospheric correction, seamless mosaicking was performed using the mosaic tool, setting the background value to zero to remove background noise. The cubio convolution method was employed for high-precision seamless stitching, ensuring spatial consistency and data integrity of the images. Subsequently, digital elevation model (DEM) data was modeled using ArcGIS Pro 3.0.1, applying the "vertical exaggeration" feature to enhance terrain details and setting lighting effects to generate shadowing, further improving the visual quality of the terrain map. The generated 3-D-DEM image was then overlaid with vector data from the study area to obtain a detailed 3-D terrain view. In addition, DJI Terra software was used to process visible remote sensing imagery from drones, including data import, quality inspection, geometric correction, image alignment and stitching, cropping, color correction, and enhancement. This ensured spatial accuracy and improved visual quality, resulting in high-quality orthophotos. The data preprocessing flowchart is shown in Fig. 2.

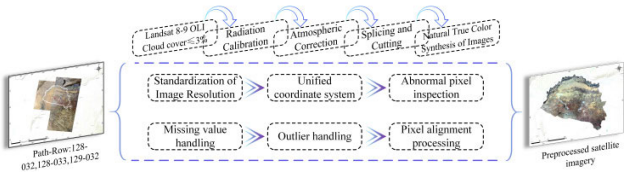


Fig. 2. Data preprocessing process.

3) Desertification Monitoring Methodological Process: Both satellite and RAV remote sensing are basic Earth observation systems used to monitor surface dynamics. The advantage of satellite remote sensing is that it can provide wide-area and multitemporal data; however, its image accuracy is difficult to adapt to fine vegetation extraction and is easily affected by cloud cover and atmospheric radiation [41]. On the other hand, RAV remote sensing can realize the acquisition of centimeter-level ultrahigh-resolution images, which improves the accuracy and compensates for the image shortcomings that are often not available owing to cloud cover in satellite remote sensing [42], [43]. However, owing to the limited operating range of RAVs, they are not applicable for wide-area remote sensing monitoring. Therefore, this study was based on the powerful effect of satellite and RAV synergy in monitoring and classifying land desertification hazards in the study area in a refined manner. First, the NDVI-albedo feature space was constructed based on Landsat 8 satellite remote sensing images of the region to calculate the DDI. Using this index, the degree of desertification was graded and the area with the highest degree of desertification was identified as the key research object. Subsequently, RAV optical images of the region were collected through field surveys, validation sample squares were established based on random sample points set in advance, and

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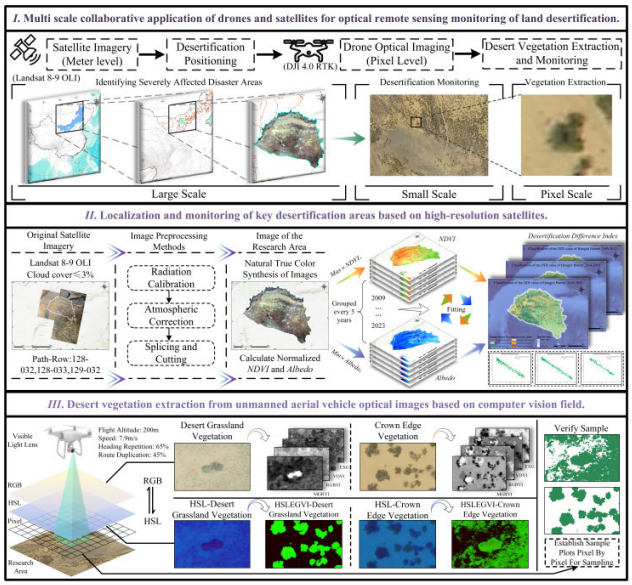


Fig. 3. Desertification monitoring methodological process.

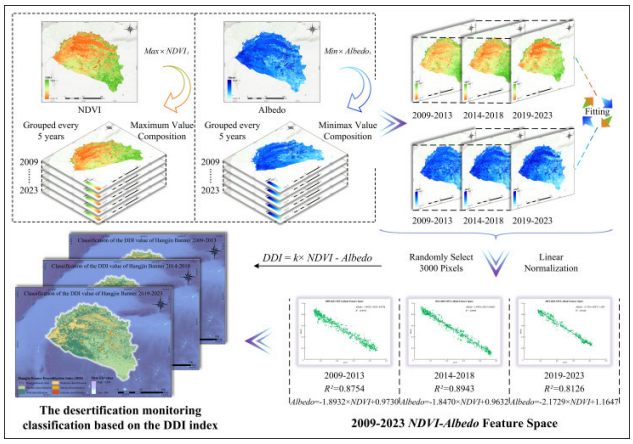


Fig. 4. Flowchart for monitoring desertification by satellite remote sensing.

the feature attribute data within the sampling squares were counted and vectorized for validation.

After acquiring the RAV optical images, the desert vegetation in the study area was refined using the vegetation index method. Because the traditional visible vegetation index is constructed based on the RGB color space, the lack of mutual independence between the three color channels, and the color information is interwoven with the luminance information in the same color space, necessitating the modification of all three channels simultaneously when adjusting the luminance or contrast of the image, making it difficult to analyze quantitatively [44], [45]. This interconnectedness between channels not only increases the risk of color distortion but also makes image-processing algorithms more complex. Therefore, in this study, HSLGEVI was constructed based on HSL color space using a tone-tuning algorithm and channel mixing enhancement technique to break through the limitations of RGB color space and make it suitable for fine extraction of desert vegetation. Finally, supervised classification of each vegetation index is performed using a support vector machine (SVM) algorithm and accuracy evaluation is completed. The specific process is illustrated in Fig. 3.

C. Wide-Area Desertification Monitoring and Classification Based on Satellite Remote Sensing

Modern remote sensing systems rely on advanced multisensor configurations, enabling satellites to simultaneously acquire higher resolution panchromatic band data as well as multispectral data and generate higher resolution multispectral images through efficient data fusion techniques. This process not only improves the level of detail of the images but also significantly enhances the ability to analyze environmental information, and the short revisit cycle of the satellite makes it possible to monitor the high-frequency dynamics of desertification and update the data in a timely manner. In addition, high-resolution images can show the textural characteristics of the ground surface in detail and accurately characterize the complex composition of feature attributes in the study area. Although high-resolution imagery may be subject to the instability of spectral statistical features because of atmospheric radiation and cloudcover, it still provides a solid database for the detailed analysis of environmental information and plays an irreplaceable role in desertification monitoring and assessment. Therefore, this study used Landsat 8 satellite data to monitor desertification trends in Hangjin Banner, and the detailed technical flowchart is presented in Fig. 4.

1) NDVI Calculation Using the Maximum Value Synthesis Method: Vegetation is important in terrestrial ecosystems, not only reflecting the ecological status of a specific region but also playing a key role in water cycle regulation, climate change response, and ecological balance maintenance [46], [47]. As an effective indicator for assessing vegetation growth and ecological changes [48], [49], the NDVI is widely used in research fields such as desertification monitoring [50], vegetation cover monitoring [51], soil and water conservation [52], and land cover classification [53]. Its value range is $[-1, 1]$, where positive values indicate the area of vegetation cover; negative or near-zero values usually indicate areas with no vegetation cover, such as bare soil, water bodies, or buildings [54]. The following equation shows the NDVI construction method:

$$\mathrm{NDVI} = \frac{N_{\mathrm{IR}} - R}{N_{\mathrm{IR}} + R}. \tag{1}$$

Because the annual maximum NDVI value can accurately capture the most vigorous stage of vegetation growth and reflect the surface cover conditions, it can also effectively eliminate the interference of atmospheric conditions, cloud cover, and solar altitude angle, thereby reducing the influence of noise on image quality. Therefore, in this study, the annual maximum value synthesis (MVC) method [55] was used to synthesize the maximum value of the NDVI data from 2009 to 2023 with a cycle of every five years, and long time series of NDVI data for the time periods 2009–2013, 2014–2018, and 2019–2023 were obtained. Among them, choosing a five-year period effectively smooths out seasonal fluctuations and short-term climate variations (such as differences between drought and wet years), thus

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providing a more accurate reflection of the long-term desertification trend. Unlike short-term data, which may be influenced by anomalous climate conditions or disasters, the five-year period helps eliminate these unstable factors. Compared to long-term data, the five-year time scale is appropriate as it captures significant changes in desertification without being overly detailed, thereby avoiding the omission of gradually developing desertification phenomena. The calculation for this is shown as follows:

$$\mathrm{NDVI}_{\max,t} = \mathrm{Max}\{\mathrm{NDVI}_{t1}, \mathrm{NDVI}_{t2}, \ldots, \mathrm{NDVI}_{tn}\} \quad (2)$$

where t denotes the time series synthesized every five years and $t_1 - t_n$ denotes the years in the time series, with t_1 being the smallest year in the time series and t_n the largest year in the time series. $\mathrm{NDVI}_{\max,t}$ denotes the maximum NDVI value synthesized for each five-year cycle, and NDVI_{t_n} denotes the NDVI observations for all years.

2) Calculation of Surface Reflectance (Albedo): Albedo is a key physical parameter of the radiant energy balance at the surface [56], [57], studies have shown that at low latitudes, surface albedo is mainly influenced by a combination of various surface characteristics, such as land-use type and topography [58], [59]. On the other hand, at high latitudes, vegetation, snow, and ice cover are the key factors leading to changes in surface albedo [60], [61]. Therefore, in this article, the inverse model developed by Liang et al. [62] was used to estimate the surface albedo in the study area, and its calculation formula is shown as follows:

$$\mathrm{Albedo} = \sum_{i=1}^{n} W_i \cdot \rho_i - 0.018 \quad (3)$$

where W_i denotes the weighting coefficient of each Landsat band, ρ_i denotes the surface reflectance of the corresponding band, i is the serial number of the band, and n is the number of bands calculated. The values of W_i and ρ_i and their corresponding values for the Landsat TM images used in this study are shown in Table II, whereas those of W_i and ρ_i and their corresponding values for the satellite images are shown in Table III.

After obtaining albedo data for the period 2009–2023 through calculations, the data were synthesized in five-year intervals, resulting in three time series of albedo data, 2009–2013, 2014–2018, and 2019–2023.

3) Normalization Process: To ensure unity of magnitude between NDVI and albedo in subsequent analyses, they were normalized to a range of values between 0 and 1 after completing the calculations. Their calculation formulas are shown in (4) and (5)

$$N = \frac{\mathrm{NDVI} - \mathrm{NDVI}_{\min}}{\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min}} \quad (4)$$

where N represents the vegetation index after normalization, and $\mathrm{NDVI}_{\min}$ and $\mathrm{NDVI}_{\max}$ represent the maximum and minimum values, respectively,

$$A = \frac{\mathrm{Albedo} - \mathrm{Albedo}_{\min}}{\mathrm{Albedo}_{\max} - \mathrm{Albedo}_{\min}}. \quad (5)$$

TABLE II
RANGE OF VALUES OF COEFFICIENTS AND THEIR CORRESPONDING VALUES OF LANDSAT TM IMAGES

ρ_i	W_i	**Spectral (μm)**	**Spatial resolutions (m)**
Red band (Band 3)	0.130	0.63–0.69	30
Blue band (Band 1)	0.356	0.45-0.51	30
Near-infrared band (Band 4)	0.373	0.76–0.90	30
SWIB1 (Band 5)	0.085	1.55–1.75	30
SWIB2 (Band 7)	0.072	2.08–2.35	30

TABLE III
RANGE OF COEFFICIENT VALUES AND THEIR CORRESPONDING VALUES OF LANDSAT 8 OLI IMAGERY

ρ_i	W_i	**Spectral (μm)**	**Spatial resolutions (m)**
Red band (Band 4)	0.130	0.64–0.67	30
Blue band (Band 2)	0.356	0.45-0.51	30
Near-infrared band (Band 5)	0.373	0.85–0.88	30
SWIB1 (Band 6)	0.085	1.57–1.65	30
SWIB2 (Band 7)	0.072	2.11–2.29	30

In (5), A represents the surface reflectance after normalization, and $\mathrm{Albedo}_{\min}$ and $\mathrm{Albedo}_{\max}$ represent the maximum and minimum values, respectively.

4) Construction of NDVI-Albedo Feature Space: In desertification monitoring, NDVI and albedo effectively reflect the desertification process from different perspectives. NDVI primarily reflects vegetation cover and health status, and is sensitive to vegetation growth, degradation, and seasonal changes, making it suitable for monitoring vegetation scarcity and the progression of desertification. On the other hand, albedo represents the surface reflectivity, with exposed soil and desertified areas typically exhibiting higher albedo values, which can indicate the degree of land degradation. By combining these two features through the NDVI-albedo feature space, it is possible to analyze the relationship between vegetation changes and surface reflectivity characteristics, effectively overcoming the limitations of using a single feature. For example, low NDVI and high albedo values typically represent areas with severe desertification, while high NDVI and low albedo values may indicate healthy ecosystems with abundant

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vegetation. By fitting these two features, constructing the NDVI-albedo feature space, and analyzing its spatiotemporal variations, desertification dynamics can be more accurately monitored, revealing the mechanisms of land degradation and restoration.

In this article, we randomly selected 3000 sample points using the "Create Fishnet" tool in ArcMap 10.8. When selecting the sample points, we applied a spatially uniform distribution method using the grid from the "Fishnet Tool" to ensure that the sample points covered all key areas of the study region, avoiding concentration in any specific area. As a result, these 3000 sample points can comprehensively reflect the vegetation and surface characteristics of the study area. Subsequently, we vectorized the sample points using the "Extract Multi Values to Points" tool, exporting vector points containing NDVI and albedo attribute values. We then performed linear regression on these two datasets using Origin 2022, thereby constructing a normalized NDVI-albedo feature space for the study area during the periods of 2009–2013, 2014–2018, and 2019–2023. The corresponding fitting equations and coefficients of determination (R^2) were obtained. The general form of the fitting equations is given as follows, respectively,

$$\text{Albedo} = a \times \text{NDVI} + b \tag{6}$$

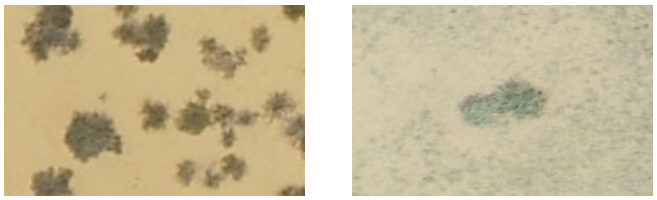
$$a \times k = -1 \tag{7}$$

where a is the coefficient of the fit equation, b is a parameter of the fit equation, and k is the DDI.

5) Constructing the DDI and Grading Desertification Levels: The DDI is a remote sensing indicator used to monitor and assess trends in desertification processes. It captures trends in vegetation cover by generating a continuous time series by differentiating the vegetation index and surface reflectance data at different points in time. This approach enables more precise identification and quantification of desertification processes and supports the dynamic analysis of environmental change. Land desertification is usually characterized by a significant reduction in ecological vegetation cover, accompanied by a significant increase in surface albedo [63]. While NDVI can reflect the surface vegetation cover, lower NDVI values usually indicate sparse or missing vegetation [64]. Albedo represents the surface reflectance characteristics, and higher albedo values are usually associated with bare or degraded land.

Therefore, in this study, based on the NDVI-albedo feature space fit with NDVI and albedo data, the DDI of the study area was calculated using (8). Subsequently, the DDI was graded using the natural intermittent point grading method [65], and five grades of potential desertification, mild desertification, moderate desertification, severe desertification, and extreme desertification were obtained [66]. Among them, the natural breakpoint classification method accurately identifies the "natural" breakpoints in the data by minimizing the within-group variance and maximizing the between-group difference [67], thus ensuring the accuracy of the DDI index classification

$$\text{DDI} = k \times \text{NDVI} - \text{Albedo}. \tag{8}$$



(a) (b)

Fig. 5. Sample study of factors influencing the accuracy of desert vegetation extraction. (a) Crown edge samples. (b) Desert meadow samples.

D. Fine-Grained Vegetation Extraction Based on RAV Visible Imagery

1) Analysis of Factors Influencing the Accuracy of Vegetation Extraction by RAV: In this article, we aim to make full use of the strong synergistic effect of satellite remote sensing image data and RAV visible remote sensing data to monitor the desertification in the study area in a refined way through "multiscale interpretation" and "data fusion application." First, we analyzed the complex environmental problems in the desert region by combining the characteristics of the visible vegetation index and systematically evaluating various factors affecting extraction accuracy. The traditional visible light vegetation index is often applied for the extraction of large-scale green vegetation from urban and forest areas. In contrast, the vegetation in the desert Gobi region is mostly coniferous shrubs, whose leaves are less green, and whose canopy range is not clear. In addition, extreme lighting conditions in this region can lead to overexposure or loss of highlight details in the captured visible images, deviation of details owing to high contrast between strong light and shadows, color distortion owing to reflected light and high light intensity, and deeper shadow textures covering the vegetation. The above problems will have a greater impact on vegetation extraction and the definition of the area covered, and seriously affect the accuracy of desertification vegetation extraction.

2) Constructing a Research Sample: After identifying the influencing factors, the RAV images were processed further. Because coniferous shrubs mainly grow in desert areas, the foliage at the edge of the canopy of these vegetation types is sparse and the boundaries are unclear. In addition, desert meadow vegetation can suffer from severe color distortion and detail loss under light conditions. All these factors have a large impact on the extraction accuracy; hence, the canopy edge and desert meadow samples were divided by enhancing the relative weights of the influencing factors (canopy and meadow vegetation) in the samples (more than 60% of the total area) to avoid the problem of false accuracy because of the low weights of the imaging factors and to improve the representativeness of the study samples. In addition, constructing these two types of samples can more effectively verify the performance of different vegetation indices in extracting vegetation from desert environments. The canopy edge samples and desert meadows are shown in Fig. 5.

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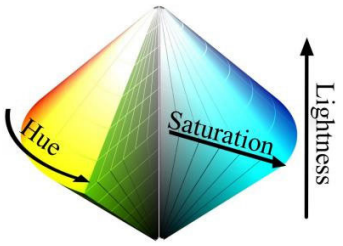


Fig. 6. HSL color model.

3) Construction of the HSLGEVI and Experiment Control Groups:

a) HSL color space: To solve the problem that the channels in the RGB color model are highly correlated and cannot be analyzed quantitatively, this study constructed a new vegetation index through the hue, saturation, and luminance (HSL) color model. The HSL color model is a widely used color representation model that provides a more intuitive way of describing colors than the RGB color space. In the HSL model, color consists of three components: hue, which indicates the type of color and is measured as an angle (0°–360°) on the color wheel; saturation, which indicates the purity or intensity of a color and ranges from 0% (gray) to 100% (fully saturated colors); and lightness, which defines the brightness of a color. The HSL color space model, which ranges from 0% (black) to 100% (white). The hue, saturation, and lightness components of the HSL color space are highly correlated with the spectral characteristics of vegetation, allowing for more accurate extraction of vegetation's green features. Moreover, compared to other color spaces, the HSL color space better meets the needs of computer vision applications, offering a more intuitive representation of colors. In addition, the HSL color space has higher computational efficiency, and the conversion process from RGB to HSL is simpler, making it particularly well-suited for large-scale data processing in remote sensing imagery. Fig. 6 shows the HSL color space model.

b) HSLGEVI construction method: This research involved the development of the HSLGEVI utilizing the HSL color space model. This was accomplished through the conversion of the RGB color model to the HSL color model, the relative independence of each channel was enhanced to solve the problem in which the channels in the RGB color model are highly correlated and cannot be quantitatively analyzed to realize fine extraction of desert vegetation. The specific construction method is as follows: first, the canopy edge and desert meadow samples were converted from the RGB color model to the HSL color model. The conversion formulas are shown as follows, which are the RGB linear normalization and detailed transformation process from the RGB color model to the HSL color model, respectively.

Linear Normalization:

$$\begin{aligned} &\mathrm{R}' = \frac{\mathrm{R}}{255} \quad \mathrm{G}' = \frac{\mathrm{G}}{255} \quad \mathrm{B}' = \frac{\mathrm{B}}{255} \\ &C_{\max} = \max(\mathrm{R}', \mathrm{G}', \mathrm{B}') \\ &C_{\min} = \min(\mathrm{R}', \mathrm{G}', \mathrm{B}') \end{aligned} \tag{9}$$

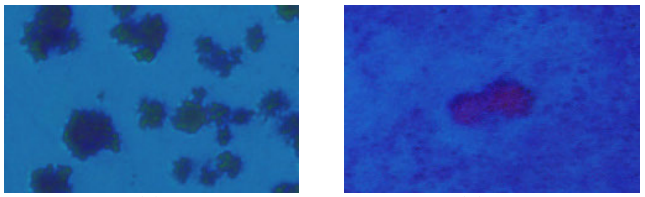


Fig. 7. HSLVI model. (a) Crown edge samples. (b) Desert meadow samples.

Hue:

$$h = \begin{cases} 0^{\circ}, & \text{if } C_{\max} = C_{\min} \\ 60^{\circ} \times \dfrac{\mathrm{G}' - \mathrm{B}'}{C_{\max} + C_{\min}} + 90^{\circ}, & \text{if } C_{\max} = \mathrm{R}' \\ & \text{and } \mathrm{G}' \geq \mathrm{B}' \\ 60^{\circ} \times \dfrac{\mathrm{R}' - \mathrm{G}'}{C_{\max} + C_{\min}} + 120^{\circ}, & \text{if } C_{\max} = \mathrm{G}' \\ 60^{\circ} \times \dfrac{\mathrm{R}' - \mathrm{G}'}{C_{\max} + C_{\min}} + 240^{\circ}, & \text{if } C_{\max} = \mathrm{B}' \\ 60^{\circ} \times \dfrac{\mathrm{G}' - \mathrm{B}'}{C_{\max} + C_{\min}} + 360^{\circ}, & \text{if } C_{\max} = \mathrm{R}' \\ & \text{and } \mathrm{G}' < \mathrm{B}'. \end{cases} \tag{10}$$

Saturation:

$$S = \begin{cases} 0, & \text{if } l = 0 \\ & \text{or } C_{\max} = C_{\min} \\ \dfrac{C_{\max} - C_{\min}}{C_{\max} + C_{\min}} = \dfrac{C_{\max} - C_{\min}}{2l}, & \text{if } 0 < l \leq \dfrac{1}{2} \\ \dfrac{C_{\max} - C_{\min}}{2 - (C_{\max} + C_{\min})} = \dfrac{C_{\max} - C_{\min}}{2 - 2l}, & \text{if } l > \dfrac{1}{2}. \end{cases} \tag{11}$$

Lightness:

$$L = \frac{C_{\max} + C_{\min}}{2} \tag{12}$$

where R, G, and B are the input visible light color values, R', G', and B' are the normalized RGB values, $C_{\max}$ represents the maximum value between the group, and $C_{\min}$ represents the minimum value between the group.

After converting the RGB color model to the HSL color model, the HSLVI was initially obtained at this time, where the HSLVI is shown in Fig. 7.

Subsequently, the saturation and brightness components were adjusted to 1.5 times their original values (it was experimentally verified that the best effect was achieved at 1.5 times). Through the hue optimization algorithm, the image was processed with partial color to enhance the green component in the dark part of the image and reduce the yellow component in the bright part. This was done to improve the sensitivity of the vegetation index model to the response of the green vegetation as well as to optimize the contrast between the vegetation and bare soil. The specific calculations are

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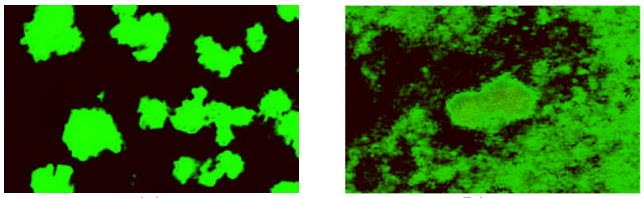


Fig. 8. HSLGEVI model. (a) Crown edge samples. (b) Desert meadow samples.

expressed in (13)–(15)

$$S_{\text{adj}} = \text{Min}\left(1, 1.5 \times S_{\text{orig}}\right)$$
$$L_{\text{adj}} = \text{Min}\left(1, 1.5 \times L_{\text{orig}}\right). \tag{13}$$

Tone Optimization Algorithm:

$$\begin{aligned} H_{\text{adj}}(x,y) &= H_{\text{HSL}}(x,y) + \Delta H_{\text{dark}} \times T_{\text{dark}}(L_{\text{HSL}}(x,y)) \\ &\quad - \Delta H_{\text{bright}} \times T_{\text{bright}}(L_{\text{HSL}}(x,y)) \\ T_{\text{dark}}(L) &= \frac{1}{1+e^{-\alpha_{\text{dark}}(L-\beta_{\text{dark}})}} \times \left(1-\gamma_{\text{dark}} \times e^{-\delta_{\text{dark}}(L-\lambda_{\text{dark}})}\right) \\ T_{\text{bright}}(L) &= \frac{1}{1+e^{-\alpha_{\text{bright}}(L-\beta_{\text{bright}})}} \\ &\quad \times \left(1-\gamma_{\text{bright}} \times e^{-\delta_{\text{dark}}(L-\lambda_{\text{bright}})}\right) \end{aligned} \tag{14}$$

$$H_{\text{adj}}(x,y) = \begin{cases} H_{\text{HSL}}(x,y) + \Delta H_{\text{dark}} \times T_{\text{dark}}(L_{\text{HSL}}(x,y)) \\ (\text{if } L_{\text{HSL}}(x,y) < L_{\text{threshold}}) \\ H_{\text{HSL}}(x,y) - \Delta H_{\text{dark}} \times T_{\text{dark}}(L_{\text{HSL}}(x,y)) \\ (\text{if } L_{\text{HSL}}(x,y) \geq L_{\text{threshold}}) \end{cases} \tag{15}$$

where the initial saturation lightness is S_{orig} and L_{orig}, and $\text{Min}(1, X)$ is used to ensure that the adjusted component values do not exceed the maximum value of HSL color space to avoid going out of range. $H_{\text{HSL}}(x, y)$ is the original hue and $H_{\text{adj}}(x, y)$ is the adjusted hue. ΔH_{dark} and ΔH_{bright} are the adjustment coefficients for darker green and brighter yellow, respectively. α_{dark} and α_{bright} control the slopes of the curves and strength of the adjustments. β_{dark} and β_{bright} are the lightness thresholds used to determine the starting point for the adjustments, γ_{dark} and γ_{bright} regulate the amplitude of the adjustment function, and δ_{dark} and δ_{bright} affect the decay speed of the adjustment function. λ_{dark} and λ_{bright} are the adjusted offsets for precise control and $L_{\text{threshold}}$ is the dividing lightness value used to distinguish between the dark and light adjustment areas. Finally, after completing the hue-tuning bias processing, the HSLGEVI was constructed, as shown in Fig. 8.

c) Construction of vegetation indices for the experiment control group: Vegetation indices are constructed by mathematically combining the reflectance of inclusions in multiple wavelength ranges with the aim of highlighting the unique spectral characteristics and details of vegetation, thereby enabling a successful differentiation between vegetative and nonvegetative elements [68], [69]. In the visible spectral range of the RGB color space, the spectral reflectance characteristics of healthy green vegetation showed that the reflectance of the green band was significantly higher than that of the red and blue bands, indicating that the absorption intensity of red and blue light by green vegetation was much greater than that of green light [70], [71]. Therefore, the current research focused on emphasizing the reflective properties of the green band to effectively distinguish vegetation from other features for vegetation extraction and to optimize the accuracy and reliability of vegetation extraction.

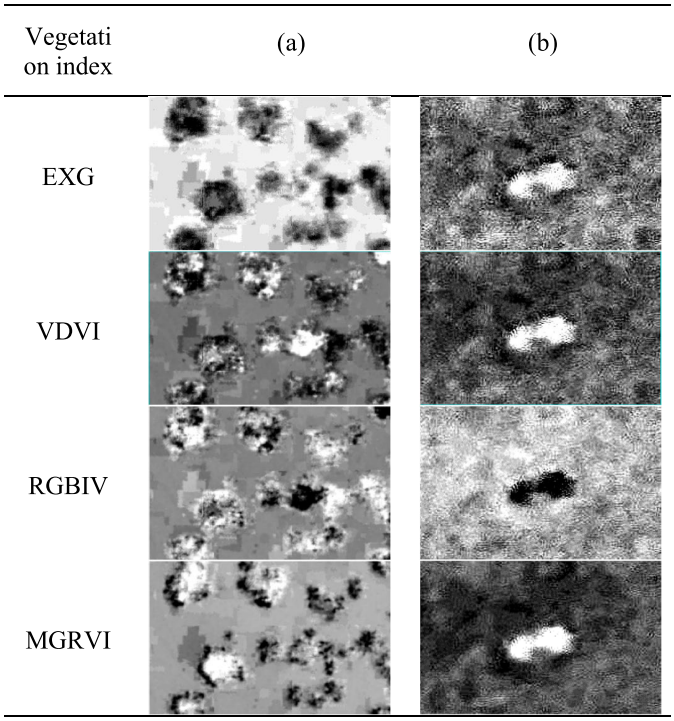


Fig. 9. Visible vegetation index model. (a) Crown edge samples. (b) Desert meadow samples.

At present, over 100 models of vegetation indices exist in the scientific literature, however, only a limited number have undergone systematic validation and gained widespread application [72], [73]. In this article, four vegetation indices that have been widely used and validated, EXG [74], vegetation difference vegetation index (VDVI) [75], RGBVI [76], and MGRVI [77], were selected as control experiments in order to compare the performance of vegetation indices constructed based on the RGB color model and HSLGEVI in desert environments. The specific construction parameters are shown as follows:

$$\text{EXG} \quad 2\rho_G - \rho_B - \rho_R \tag{16}$$
$$\text{VDVI} \quad \frac{2\rho_G - \rho_B - \rho_R}{2\rho_G + \rho_B + \rho_R} \tag{17}$$
$$\text{RGBVI} \quad \frac{\rho_B + \rho_R}{2\rho_G} \tag{18}$$
$$\text{MGRVI} \quad \frac{\rho_G^2 - \rho_R^2}{\rho_G^2 + \rho_R^2}. \tag{19}$$

In this formula, ρ_R, ρ_G, and ρ_B represents the red, green, and blue bands. The model for each vegetation index, obtained based on the above equations, is shown in Fig. 9.

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E. Supervised Classification Based on Each Vegetation Index

After calculating the HSLVI, HSLGEVI, and the vegetation index from the RGB-based control experiment, this study applied the SVM algorithm to define the "vegetation" and "bare soil" regions of interest for supervised classification in both study samples. The Jeffries–Matusita distance [78], ranging from 0 to 2, is commonly used to assess separability between categories. A higher value indicates better separation and stronger discrimination between regions, with a value above 1.8 deemed acceptable and above 1.9 considered accurate [79]. To ensure classification accuracy, the separability between different regions in the training samples was kept above 1.9. After determining the regions of interest, the original and desert meadow samples of the canopy margin were supervised and classified using an SVM.

1) SVM Algorithm: Originally introduced by Vladimir Vapnik and Alexey Chervonenkis [80], [81], [82] the SVM is a traditional machine learning algorithm grounded in statistical theory. It is extensively applied in both classification and regression analysis [83]. The fundamental concept involves maximizing the margin between distinct categories by identifying an optimal hyperplane within a high-dimensional space to fulfill a classification objective. To determine the optimal hyperplane, the selection criterion maximizes the interval between classes, that is, the distance from the support vector (i.e., the closest sample point to the hyperplane) to the hyperplane [84], [85]. This approach allows SVMs to have excellent generalization ability on training data and perform well on unknown data. Among them, the kernel function is a key tool for SVM to deal with nonlinear problems. It converts difficult nonlinear issues into linear forms by translating the data from the initial feature space to a feature space with higher dimensions [86], [87]. Common types of kernel functions consist of the linear kernel, which is suitable for problems that can be solved linearly; the polynomial kernel, designed to capture polynomial relationships among features; the radial basis function (RBF) kernel, known for its ability to handle intricate nonlinear relationships and is widely utilized in SVMs; and the sigmoid kernel, akin to the activation function found in neural networks [88]. In addition, the performance of SVM models is affected by several hyperparameters, such as the penalty parameter C, which is used to control the tradeoff between the complexity of the classifier and training error [89], [90]. A larger value of C results in a classifier with a smaller error in the training set but may lead to overfitting, whereas a smaller value of C increases the training error but helps to improve the generalization of the model.

2) SVM Performance Advantages: SVMs have certain advantages in recognizing small samples of visible vegetation. First, SVMs can effectively cope with common high-dimensional data structures in vegetation images by virtue of their superior processing capability in high-dimensional feature spaces. Second, SVMs optimize the decision boundary by maximizing the category interval, which significantly improves the generalization ability of the models and is especially crucial for accurate classification in small-sample situations. In addition, the kernel techniques of SVMs (e.g., RBF and polynomial kernel functions) can map the data into a high-dimensional space and effectively deal with complex nonlinear classification problems [91], [92], thus improving the ability to distinguish between different vegetation types and define the edges of the vegetation canopy. Compared with other algorithms, such as the k-nearest neighbor algorithms, SVMs show stronger robustness in coping with data noise and category imbalance, and their hyperparameter optimization ability enables them to be precisely adjusted to fit specific tasks. These characteristics give SVMs superior performance and reliability in vegetation recognition with small sample data. Fig. 10 illustrates the SVM model.

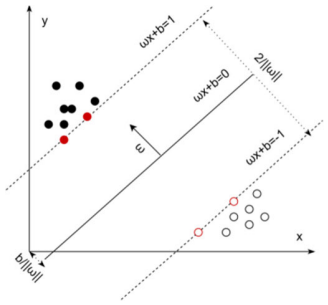


Fig. 10. SVM model.

F. Precision Evaluation

1) Constructing Validation Samples: In this study, data were collected through field surveys, vectorized, and constructed into validation samples. After determining the study area of the canopy edge and desert meadow samples, the entire study samples were divided into grids of images in terms of image elements by using the "Create Fishing Grid" tool of Arcgis 10.8. Field data were collected, and the spatial resolution of the RAV images was used to calculate the actual area occupied by each pixel. Sample squares were constructed based on the pixel grid, and the attributes of the features that accounted for more than 60% of the orthorectified projections within the sample squares were regarded as the attributes of the sample squares. After all the sample data were counted, they were vectorized to construct a validation sample, and the outliers in the constructed validation sample were calibrated by visual interpretation. Finally, the classification results (training samples) and validation samples are supervised to construct a confusion matrix and complete the accuracy evaluation.

2) Constructing a Confusion Matrix: The confusion matrix serves as an essential instrument for evaluating the classification accuracy of remote sensing images, offering a structured comparison between the predicted and actual categories generated by the classification model [93], [94]. This matrix is presented in the form of a 2-D table, where the rows denote the actual categories and the columns indicate the predicted categories [95], [96], [97]. Utilizing the confusion matrix, one can compute performance metrics including overall classification accuracy, producer accuracy, user accuracy, $F1$ score, and the kappa coefficient. These metrics help to deeply analyze the performance of the classification model in various categories and identify misclassification patterns, thus providing data

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support for further optimization of the model. The constructed confusion matrix is presented in the Appendix.

3) Accuracy Evaluation Based on the Confusion Matrix:

1) The overall accuracy of classification refers to the ratio of accurately identified samples to the complete count of actual samples [98]. In the classification task, this metric reflects the correct classification rate of the model on the whole dataset, thus measuring the average performance of the model on all categories, which is calculated as shown as follows:

$$\mathrm{OA} = \frac{\sum_{i=1}^{k} X_{ii}}{N} \tag{20}$$

where N is the total number of samples, k is the total number of categories, and X_{ii} is the number of correctly categorized samples.

2) Producer accuracy refers to the percentage of actual samples from a specific category that are correctly identified as belonging to that category in a classification task [99]. It measures the performance of a classification model in recognizing a particular category, reflecting the proportion of all samples that belong to that category and that are correctly identified. It is used to assess the model's ability to detect each category and is a key measure of the validity and reliability of classification models on a specific category. It is calculated as shown as follows:

$$\mathrm{PA}_i = \frac{X_{ii}}{X_{+i}} \tag{21}$$

where X_{+i} is the number of samples in category i.

3) In classification tasks, user accuracy is defined as the probability of a particular category being correctly recognized by a classifier [100]. It is used to assess the ability of a classification model to accurately assign image elements to a particular category and measure the proportion of image elements labeled as belonging to a particular category in the classification results that actually belong to that category, reflecting the accuracy and reliability of the classifier in a particular category. Its calculation is shown as follows:

$$\mathrm{UA}_i = \frac{X_{ii}}{X_{i+}}. \tag{22}$$

In these equations, N represents the total reference pixels, and k denotes the total number of rows or columns in the confusion matrix. X_{ii} refers to the number of image elements correctly classified into a specific category on the main diagonal of the confusion matrix. X_{i+} is the number of samples predicted to be in category i.

4) The $F1$ score is used to evaluate the overall performance of a classification model in terms of precision and recall. It is the harmonic mean of precision and recall, with a range from 0 to 1. The closer the score is to 1, the better the model's performance. The $F1$ score is especially useful when dealing with imbalanced classes, as it strikes a balance between precision and recall, offering a more comprehensive assessment of the model. The specific calculation method is shown as follows:

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{23}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{24}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{25}$$

where precision refers to the proportion of actual positive samples among those predicted as positive by the model; recall refers to the proportion of actual positive samples correctly predicted as positive by the model; true positive (TP) is the number of samples correctly predicted as positive by the model; false positive (FP) is the number of samples incorrectly predicted as positive by the model; and false negative (FN) is the number of samples incorrectly predicted as negative by the model.

5) The kappa coefficient serves as a statistical measure that effectively evaluates how well a classification model performs. It assesses the concordance between the predictions generated by the model and the actual outcomes, while also adjusting for the impact of random agreement [101]. The range of its values spans from -1 to 1, with higher values reflecting improved accuracy and trustworthiness of the classification model. The following equation presents the formula used for this calculation:

$$K = \frac{N \times \sum_{i=1}^{n} P_{ii} - \sum (P_{i+} \times P_{+i})}{N^2 - \sum (P_{i+} \times P_{+i})}. \tag{26}$$

In this context, K represents the kappa coefficient, while P_{ii} denotes the count of image elements for a specific category located on the main diagonal of the confusion matrix (which reflects the accurately categorized image elements). N indicates the total number of rows present in the error matrix, P_{i+} refers to the aggregate of the ith row, P_{+i} signifies the total for the ith column, and n represents the overall count of reference image elements.

III. Results

A. Results of Monitoring Desertification Trends in the Hangjin Banner Region

1) Linear Normalization Equations and Their Determination Coefficients in NDVI-Albedo Eigenspace: By fitting the NDVI and albedo indices to the linear normalization, the NDVI-albedo feature space can be constructed for the periods 2009–2013, 2014–2018, and 2019–2023. Furthermore, the linear normalization equations and coefficients of determination, R^2 values, for the NDVI-albedo feature space can be obtained, as shown in Table IV and Fig. 11.

The coefficients of determination (R^2) in the table reflect the closeness of the linear relationships between the independent and dependent variables in the NDVI-albedo feature space. The value of R^2 is usually between 0 and 1, and the closer it is to 1, the better the fit of the model.

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TABLE IV
LINEAR NORMALIZATION EQUATIONS AND COEFFICIENTS OF DETERMINATION FOR THE NORMALIZED NDVI-ALBEDO EIGENSPACE

Year	Fitted equation	R^2
2009–2013	$Albedo = -1.8932 \times NDVI + 0.9730$	0.8754
2014–2018	$Albedo = -1.8470 \times NDVI + 0.9623$	0.8943
2019–2023	$Albedo = -2.1729 \times NDVI + 1.1647$	0.8226

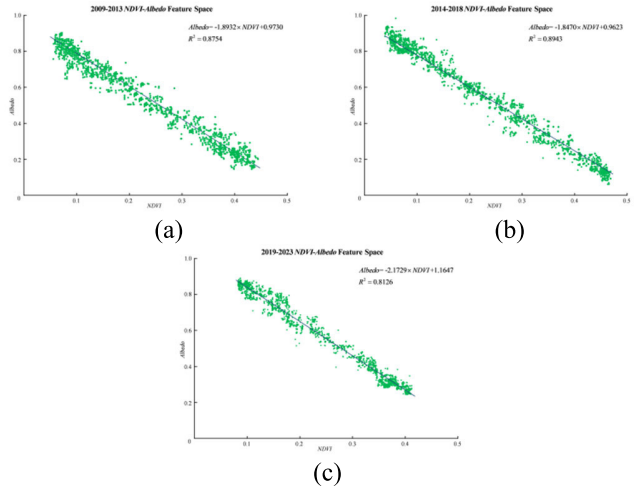


Fig. 11. Linear fitting results for the NDVI-albedo feature space. (a) 2009–2013, (b) 2014–2018, and (c) 2019–2023.

2) Classification of Desertification Based on the DDI: Based on the linear normalized equation in the NDVI-albedo feature space, the DDI of Hangjin Banner was calculated. Using the "natural discontinuity grading method," four suitable critical points in the thresholds of the DDI index of each group of years were determined. This was done to classify the desertification level into five grades. The expression of the DDI and the classification of desertification degree are shown in Tables V and VI and Fig. 12, respectively.

As shown in Fig. 12, from 2009 to 2023, the overall land desertification trend in Hangjin Banner has improved. However, the areas most severely affected by desertification are still concentrated in the western and central parts of Hangjin Banner, with a tendency for desertification to expand from the northwest to the northeast and from the central to the southeast.

3) Desertification Monitoring and Assessment Based on the DDI: This study assessed the degree of desertification based on the DDI, and determined the areas that need to be focused on monitoring by analyzing the desertification evolution trend of Hangjin Banner during the period of 2009–2023, whose spatial and temporal trends are shown in Fig. 13. The detailed data are shown in Table VII.

Fig. 13 shows the spatial and temporal characteristics of land desertification in Hangjin Banner. In the time dimension:

TABLE V
RESULTS OF THE CALCULATION OF THE DESERTIFICATION MARGIN INDEX

Year	Coefficient k	Level of desertification	DDI threshold
2009–2013	0.5282	Extreme	[-0.2334, 0.4949]
		Severe	[0.4949,0.5281]
		Moderate	[0.5281,0.5480]
		Mild	[0.5480,0.5746]
		Potential	[0.5746,0.6543]
2014–2018	0.5414	Extreme	[-0.7676, -0.0536]
		Severe	[-0.0536,-0.0090]
		Moderate	[-0.0090,0.0071]
		Mild	[0.0071,0.0274]
		Potential	[0.0274,0.6156]
2019–2023	0.4602	Extreme	[-0.7310, -0.2144]
		Severe	[-0.2144,-0.0766]
		Moderate	[-0.0766,-0.0168]
		Mild	[-0.0168,0.0287]
		Potential	[0.0287,0.5605]

TABLE VI
EXPRESSION OF THE DDI FOR 2009–2023

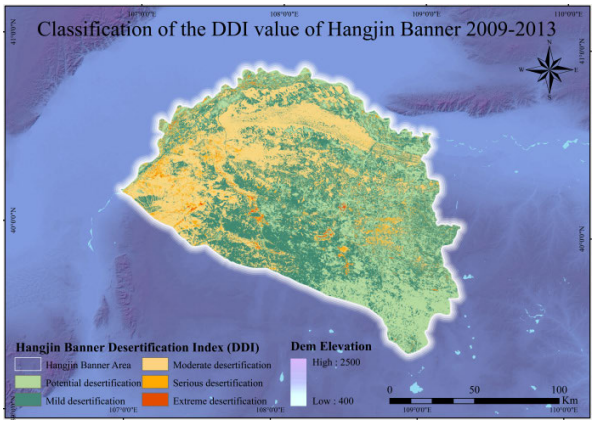
Year	DDI expression
2009-2013	$DDI = 0.5282 \times NDVI - Albedo$
2014-2018	$DDI = 0.5414 \times NDVI - Albedo$
2019-2023	$DDI = 0.4602 \times NDVI - Albedo$

TABLE VII
AREA OF EACH DESERTIFICATION CLASS

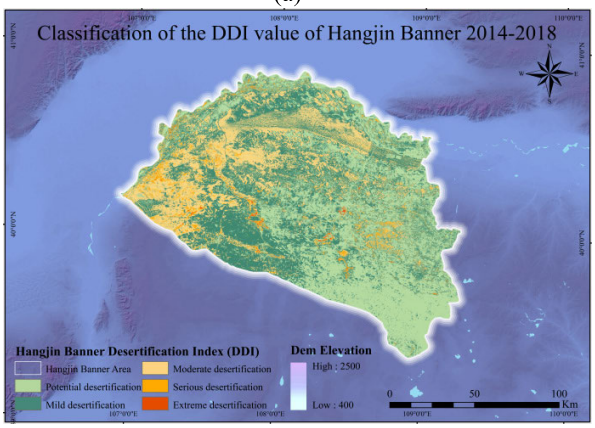
Level of desertification	Area and percentage of desertification by level/km² (Percentage)		
	2009-2013	2014–2018	2019–2023
Extreme desertification	72.98 (0.39%)	50.12 (0.27%)	55.79 (0.30%)
Severe desertification	581.23 (3.08%)	535.70 (2.84%)	150.89 (0.80%)
Moderate desertification	6400.99 (33.97%)	4479.89 (23.77%)	3962.02 (21.02%)
Mild desertification	7085.25 (37.60%)	6948.92 (36.88%)	7833.36 (41.57%)
Potential desertification	4704.89 (24.97%)	6829.83 (36.24%)	6842.43 (36.31%)

the desertification disaster in Hangjin Banner was at its peak from 2009 to 2013, during which the areas of extreme and severe desertification reached maximum values of 72.98 and 581.24 km^2, accounting for 0.38% and 3.08% of the total area, respectively. Subsequently, between 2014 and 2018, the problem of desertification was alleviated, with the area of extreme desertification reduced to 50.12 km^2 and that of severe desertification reduced to 535.70 km^2, whereas moderate desertification was effectively controlled, with its area reduced by 1921.10 km^2 compared with the past five years. During 2019–2023, the areas of light and potential

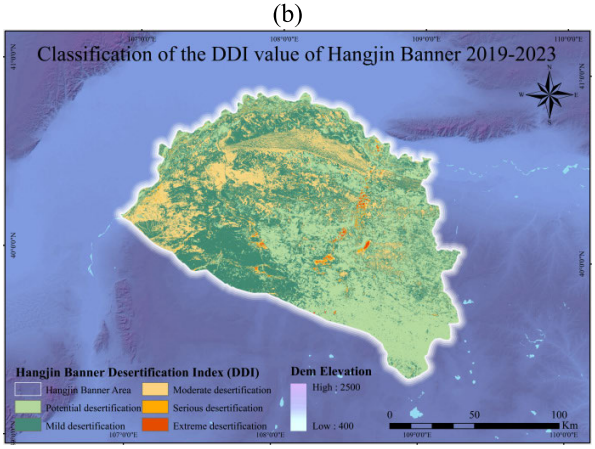
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(a)



(b)



(c)

Fig. 12. Grading of the DDI. (a) 2009–2013, (b) 2014–2018, and (c) 2019–2023.

desertification reached 7833.36 and 6842.43 km^2, accounting for 41.57% and 36.61% of the total area, respectively, and the degree of desertification was effectively improved.

From the spatial dimension based on the analysis of Dem elevation data, it can be seen that the desertification changes in Hangjin Banner in the period of 2009–2023 show obvious differences in altitude. The degree of desertification is more serious in low altitude areas (400–800 m), and during 2009–2013, mild desertification transformed to moderate, severe, and even extreme desertification, with significant ecological degradation. During 2014–2018, some areas of moderate and severe desertification recovered, but the trend of

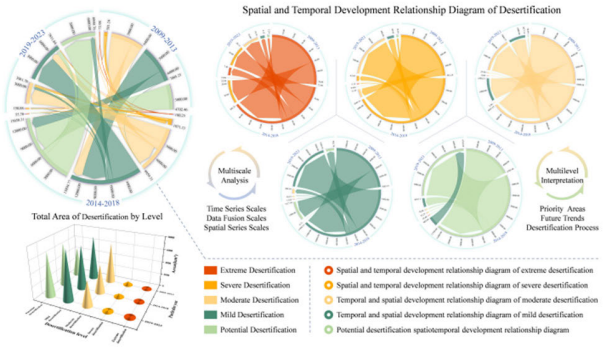


Fig. 13. Spatial and temporal trends in the desertification process in Hangjin Banner.

extreme desertification still exists to expand. By 2019–2023, the trend of desertification has improved to some extent, but the trend of land desertification expansion still exists in the future. The change of desertification in the middle altitude area (800–1500 m) is relatively gentle, mainly dominated by mild and moderate desertification, and the degree of desertification shows a certain trend of recovery and mitigation. The desertification change in the high altitude area (1500–2500 m) is the smallest, and the overall level of potential and mild desertification is maintained, and the ecosystem is relatively stable. Overall, the desertification changes in Hangjin Banner are negatively correlated with the altitude, and the desertification is more intense in the low-altitude areas, while the desertification process is more stable in the high-altitude areas. In summary, the DDI classification can effectively assess the desertification process and spatial and temporal trends in Hangjin Banner, and provide a scientific basis for identifying key monitoring areas.

B. RAV Refined Desert Vegetation Extraction Results

1) Results of Supervised Classification by Vegetation Index: In this study, we constructed "vegetation" and "bare soil" regions of interest based on each vegetation index and ensured that the separation between different regions of interest in each training sample was greater than 1.9. We used the SVM algorithm to supervise the classification of the canopy edge and desert meadow samples. The classification results are shown in Fig. 14.

Based on the analysis of Fig. 14, in the crown edge samples, the intense illumination in the desert environment caused shadow textures, leading to color distortion and spectral mixing at the edges of the vegetation canopy. This irregular spectral variation made it difficult for traditional vegetation indices, such as VDVI, RGBVI, MGRVI, and HSLVI, to accurately distinguish between the fragmented vegetation at the canopy edge and the background features, resulting in difficulty in identifying "vegetation pixels" and causing relatively severe under-classification. In the desert meadow samples, due to the low chlorophyll content in the meadow vegetation, the surface reflectance characteristics were similar to those of bare soil. Indices such as EXG, VDVI, MGRVI, and RGBVI misclassified a large number of "bare soil pixels" as "vegetation pixels," leading to serious misclassification

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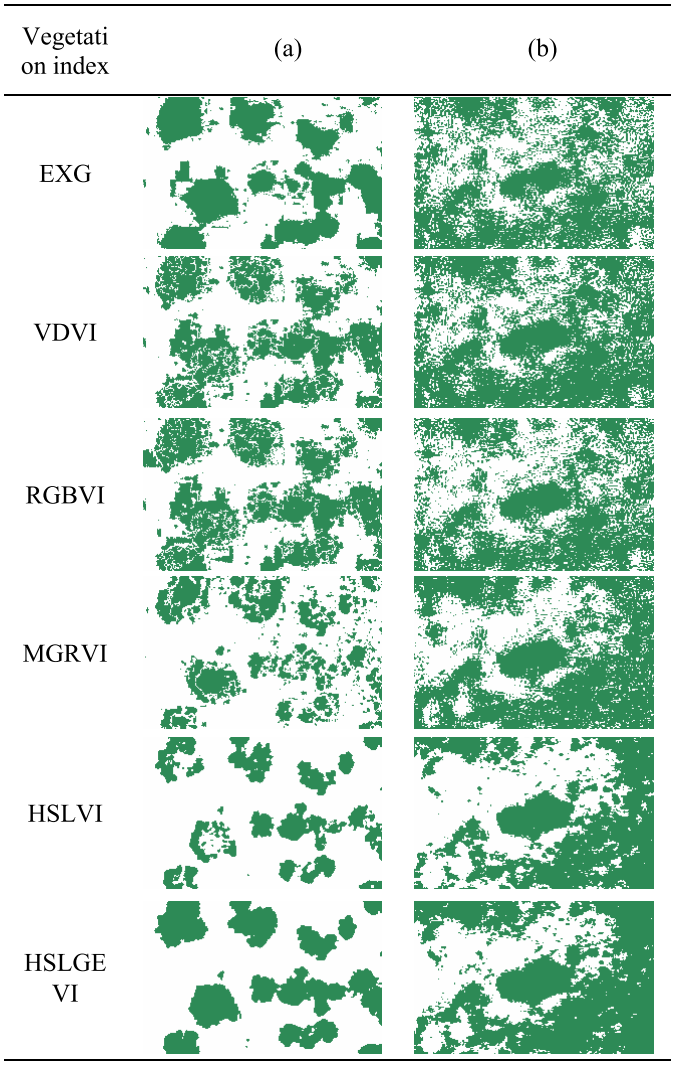


Fig. 14. Supervised classification results for each vegetation index model. (a) Crown edge samples. (b) Desert meadow samples.

issues. Subsequently, the HSLGEVI vegetation index can be constructed based on the HSL color space by utilizing color tone optimization algorithms, effectively solving this issue.

2) Evaluation of the Accuracy of Classification Results: Based on the multiple key indicators of accuracy evaluation constructed by the confusion matrix, the supervised classification results of canopy edge vegetation samples as well as desert meadow vegetation samples were evaluated in terms of accuracy, and the specific results are shown in Fig. 15 and Tables VIII and IX.

By analyzing Table VIII and Fig. 15, we can see that in the canopy edge samples, EXG and HSLVI have high producer accuracy, but the user accuracy, $F1$ score, and kappa coefficient are low. This indicates that these two vegetation indices can recognize most of the real "vegetation" pixels, but among the pixels labeled as "vegetation," the proportion of real vegetation is lower than 90%, which misclassifies some of the "bare soil" pixels. However, the proportion of real vegetation in the images labeled as "vegetation" is less than 90% and some of the "bare soil" images are misclassified as "vegetation" images, which is a serious misclassification

TABLE VIII
CANOPY EDGE VEGETATION SAMPLE ACCURACY STATISTICS

Vegetation index	Overall accuracy	User accuracy	Producer accuracy	F1 score	Kappa
EXG	87.05%	85.60%	95.13%	81.33%	0.7152
VDVI	78.53%	90.14%	77.31%	70.23%	0.5389
RGBVI	77.41%	91.05%	74.55%	69.71%	0.5251
MGRVI	82.19%	85.61%	89.14%	70.02%	0.5736
HSLVI	91.49%	89.01%	100.00%	84.13%	0.7852
HSLGEVI	**98.26%**	**98.51%**	**96.00 %**	**97.24%**	**0.9597**

TABLE IX
CANOPY EDGE VEGETATION SAMPLE ACCURACY STATISTICS

Vegetation index	Overall accuracy	User accuracy	Producer accuracy	F1 score	Kappa
EXG	66.50%	51.41%	59.98%	73.24%	0.2880
VDVI	74.33%	64.33%	58.13%	80.91%	0.4200
RGBVI	73.52%	59.49%	73.90%	78.28%	0.4468
MGRVI	80.68%	70.17%	76.96%	84.89%	0.5829
HSLVI	85.91%	71.11%	99.94%	88.32%	0.7160
HSLGEVI	**93.76%**	**87.24%**	**96.13%**	**95.21%**	**0.8660**

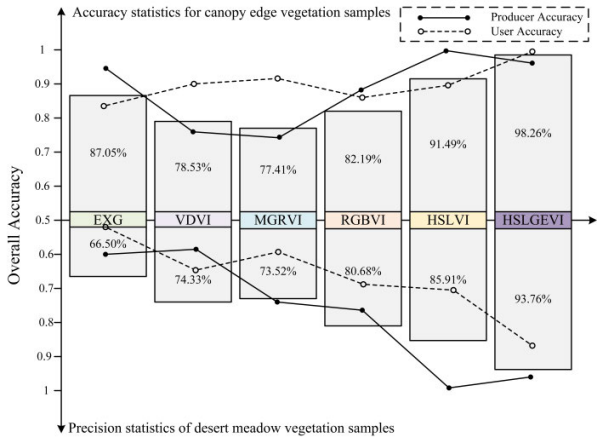


Fig. 15. Overall trend in experimental accuracy.

problem. This phenomenon suggests that defining the canopy edge of vegetation in these two vegetation indices is difficult, which can lead to large errors in the calculation of the area covered by vegetation. VDVI and RGBVI vegetation indices have high user accuracy, but the total producer accuracy, $F1$ score, and kappa coefficient are low. This indicates that in the study sample, the proportion of real "vegetation" pixels in the pixels labeled as "vegetation" is high, but some real "vegetation" pixels in the overall sample remain to be recognized. Furthermore, the problem of underclassification also exists. However, some real "vegetation" images remain to be recognized in the whole sample as well as the problem of missed classification. Table IX shows that the accuracies of the EXG, VDVI, and RGBVI in the desert meadow vegetation samples were generally lower than 75%. This is because of the high correlation between the channels in the RGB color

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space, the loss of highlight details under extreme lighting conditions, and the detailed bias and color distortion caused by the high contrast of strong light between vegetation and bare soil. Among the two experimental samples, the accuracy of the MGRVI was relatively stable, but it was difficult to meet the requirements of fine monitoring and extraction of desert vegetation. However, the overall accuracy (98.26%), producer accuracy (96.00%), user accuracy, and kappa coefficient of HSLGEVI reached more than 95%. This indicates that the classification results of the HSLGEVI are highly consistent, and the actual classification matches the expected classification. It shows superior performance in desert vegetation extraction and better copes with complex interference factors in the desert environment to realize accurate desert vegetation extraction.

IV. Discussion

A. Integration of Multisource Remote Sensing Applications Based on Strong Synergies

Both satellites and drones are important Earth observation systems used to monitor surface environmental changes [102]. While the former has more time series and a wider monitoring range, the latter has more flexibility and very high spatial resolution. Despite the potential synergies between these two data sources, they are typically used separately in disaster remote sensing applications. The strong synergistic effects of "multiscale interpretation" and "data fusion application" have not been fully utilized. In this study, we calculated the DDI to monitor the desertification process in the whole region of Hangjin Banner in a wide area and multitemporal phase and accomplished the precise positioning of the serious areas of land desertification. Subsequently, relying on the high resolution and flexibility of the RAV, vegetation information of the region was extracted in a refined manner to assess the vegetation coverage and desertification process, providing systematic scientific data support for desertification control in the region. In future research on synergistic effects, improving the accuracy and efficiency of data fusion will become a key development goal in this field. The core challenge in accuracy enhancement lies in the fine registration and information extraction of multisource remote sensing data. In recent years, deep learning technologies, particularly methods such as convolutional neural networks (CNNs) and generative adversarial networks (GANs), have shown tremendous potential in remote sensing data fusion [103], [104]. These methods can efficiently handle complex data structures, and automatically learn and extract optimal features from different data sources, thus improving fusion accuracy while reducing reliance on manual intervention. Therefore, future research could further design specialized deep learning models based on the synergistic characteristics of satellite and RAV remote sensing data to achieve more efficient and accurate automated data fusion. In addition, improving the efficiency of synergistic data fusion is equally critical. By adopting efficient computational frameworks for processing multisource remote sensing data, computational efficiency can be significantly improved [105]. Leveraging parallel computing and cloud computing platforms can accelerate the processing of large-scale remote sensing data, further optimizing the use of computational resources and effectively enhancing the overall efficiency of data fusion.

B. Application and Analysis of the DDI

In desertification monitoring, single vegetation indicators such as NDVI or FVC are typically used for ecological classification and desertification assessment [106]. However, because of the limited information dimension of these single indicators, the large influence of seasonal changes, and the cyclical fluctuation of data, significant limitations exist in assessing the desertification process and comprehensively reflecting the multidimensional characteristics and development trends of desertification is difficult [24], [107]. Therefore, in this study, based on satellite remote sensing images from 2009 to 2023, NDVI and albedo indices were calculated and processed using the MVC method to reduce cyclical fluctuation errors caused by environmental changes. Subsequently, the NDVI-albedo feature space values for 2009–2013, 2014–2018, and 2018–2023 were constructed for every five-year cycle, and their inversion accuracies reached 87.54%, 89.43%, and 81.26%, respectively. This feature space integrates the vegetation growth conditions and surface optical properties to provide a more comprehensive view of this information. Based on the NDVI-albedo feature space, this study further calculated the DDI, based on which the desertification degree of Hangjin Banner was classified into five grades, to assess desertification trends more accurately. The results show that the distribution of extremely desertified areas in Hangjin Banner is irregular, whereas the areas of severe and moderate desertification are mainly concentrated in mountainous areas in the west and north and the areas of mild and potential desertification are mainly distributed in the central plains.

C. Refinement of the HSLGEVI for Extracting Vegetation

The vegetation index is a key indicator in remote sensing for quantifying the surface vegetation cover and growth status [108]. With the continuous advancement of remote sensing technology, the application of visible vegetation index in RAV vegetation extraction has become increasingly prominent [109]. As a concise and efficient metric tool, the vegetation index not only improves the interpretation capability of RAV-visible remote sensing images but also refines the extraction of detailed information, such as vegetation type and vegetation coverage area, providing effective data support for environmental monitoring and resource management. Although some vegetation indices constructed based on the RGB color space have been widely verified and applied in RAV vegetation extraction, the inherent limitations of the RGB color space itself lead to greater limitations in the accuracy of vegetation extraction in complex desert environments. Therefore, this study constructed an HSLGEVI based on the HSL color space. Compared to the RGB color space vegetation index, the HSLGEVI significantly reduced the interdependence and correlation between channels by decomposing color into three uncorrelated components: hue, saturation, and

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brightness. This decomposition allows each color component to describe different color characteristics independently, resulting in a more focused and accurate representation of the color variety by the hue component. The hue optimization algorithm for image bias processing can enhance the green tones in the dark part of the image and reduce the yellow tones in the bright part, which greatly improves the sensitivity of the response of the HSLGEVI model to green vegetation and enhances the contrast between vegetation and bare soil. Thus, HSLGEVI can accurately capture vegetation characteristics and clearly define the edges of the vegetation canopy under high-intensity light conditions. This effectively avoids the loss of highlight details and color distortion under extreme lighting conditions and realizes the accurate extraction of shallow meadow vegetation on bare ground. This method solves the problems of misclassification and omission of RGB color space vegetation indices in complex environments and significantly improves the accuracy and robustness of vegetation extraction.

In addition to the aforementioned advantages, this vegetation index also has certain limitations. First, the HSLGEVI model has a strong dependence on high-quality and high-resolution data, requiring strict precision and detail in the input data. Furthermore, since the index is based on visible light imagery, its performance is constrained by the limitations of the visible light spectral bands. In diverse land cover environments, particularly in urban areas, there are numerous green objects whose spectral signals may closely resemble those of vegetation, thus interfering with vegetation extraction results. Therefore, in the future, it may be beneficial to incorporate near-infrared bands to increase spectral diversity, thereby enhancing the adaptability and accuracy of the HSLGEVI vegetation index model in complex land cover environments. In the future, the optimization and application of the HSLGEVI model will further focus on improving accuracy and adaptability to better cope with complex environmental conditions and data noise. Meanwhile, future research will explore its extensive application in various ecosystems, including urban green spaces, wetlands, grasslands, and polar regions, which have unique vegetation structures and ecological backgrounds, posing higher demands on the model's adaptability. In addition, the research will integrate drone-based multispectral imaging data, particularly incorporating near-infrared spectra, which can further enhance the HSLGEVI vegetation index's ability to capture vegetation information, effectively identify subtle vegetation changes, and thereby improve the accuracy and adaptability of vegetation monitoring.

V. Conclusion

The problem of land desertification in the Hangjin Banner area of Inner Mongolia, China, is becoming increasingly serious because of the combined effects of anthropogenic activities, and irrational land use, which pose a serious threat to the stability of the ecosystem and sustainable development of the region. Moreover, existing studies usually neglect the synergistic effect of satellite remote sensing and RAV remote sensing, or only focus on the stage of the weak synergistic effect of "data comparison," which does not give full play to the monitoring advantages of the two remote sensing technologies. Therefore, based on the strong synergistic effect of "multiscale interpretation" and "data fusion application" of satellite remote sensing and RAV remote sensing, this study exploits the potentials of multisource remote sensing synergistic application in desertification monitoring in terms of wide-area coverage, long time-series monitoring, multiscale analysis, and high-precision interpretation. The potential of multisource remote sensing synergistic application in desertification monitoring was deeply explored in the wide-area coverage, long time-series monitoring, multiscale analysis and high-precision interpretation, and desertification monitoring classification and refined vegetation extraction were carried out in Hangjin Banner.

In this study, NDVI and albedo data were acquired using Landsat 8 satellite remote sensing images during the period 2009–2023, and the NDVI-albedo feature space was constructed using linear normalization fitting. Based on this feature space, the DDI was calculated to classify the degree of land desertification in Hangjin Banner into five grades and analyze the spatial and temporal change characteristics. Among them, from 2009 to 2013, the desertification disaster in Hangjin Banner reached its peak, and the areas of extreme and severe desertification reached 72.98 and 581.24 km^2, respectively. From 2019 to 2023, the desertification in Hangjin Banner was effectively improved, and the areas of mild and potential desertification reached 7833.36 and 6842.43 km^2, accounting for 41.57% and 36.61% of the total area, respectively. In addition, the development of land desertification disasters in Hangjin Banner is closely related to elevation. Between 2009 and 2023, desertification was more severe in low-altitude areas, with significant ecological degradation; desertification was relatively mild in mid-altitude areas, showing some signs of recovery; and the high-altitude areas experienced the least change, with the progression of land desertification being more stable.

Wide-area multitemporal monitoring and grading of the desertification process in the Hangjin Banner region were carried out. Then, using centimeter-level high-resolution RAV remote sensing, the vegetation information of the region was extracted in a refined manner by constructing a visible vegetation index to further evaluate the land desertification process. However, because of the high correlation between the three color channels in the RGB color space, more serious detail loss and color distortion occur under high light intensity conditions in the desert, resulting in a significant decrease in extraction accuracy. Therefore, this study proposes a new vegetation index, HSLGEVI, based on the HSL color space using a hue-tuning algorithm, which shows higher accuracy and better applicability under complex lighting conditions. The results show that the HSLGEVI and other visible light vegetation indices that have not been tuned by the hue algorithm generally have poor accuracy and are unstable in the "accuracy influencing factor experiment, which makes it difficult to adapt to vegetation extraction in desert environments. However, for the two types of experimental samples, the overall classification accuracy of HSLGEVI reached 98.26% and 93.76%, with $F1$-scores of 97.24% and 95.21%, respectively. This indicates that the index effectively overcomes the limitations of the RGB

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TABLE X
CONFUSION MATRIX FOR ACCURACY EVALUATION OF CANOPY EDGE VEGETATION SAMPLES

Vegetation Index	Feature Type	Validation Sample Vegetation	Validation Sample Bare Soil	Total
EXG	Vegetation	10193	3605	13798
	Bare soil	1098	21432	22530
	Total	11291	25037	36328
VDVI	Vegetation	9173	5680	14853
	Bare soil	2118	19357	21475
	Total	11291	25037	36328
MGRVI	Vegetation	7541	2720	10261
	Bare soil	3750	22317	26067
	Total	11291	25037	36328
RGBVI	Vegetation	9456	6372	15828
	Bare soil	1835	18665	20500
	Total	11291	25037	36328
HSLVI	Vegetation	8199	0	8199
	Bare soil	3092	25037	28129
	Total	11291	25037	36328
HSLGEVI	Vegetation	11123	463	11586
	Bare soil	168	24574	24742
	Total	11291	25037	36328

TABLE XI
CONFUSION MATRIX FOR ACCURACY EVALUATION OF DESERT MEADOW VEGETATION SAMPLES

Vegetation Index	Feature Type	Validation Sample Vegetation	Validation Sample Bare Soil	Total
EXG	Vegetation	16609	5037	21646
	Bare soil	7134	7548	14682
	Total	23743	12585	36328
VDVI	Vegetation	19687	5269	24956
	Bare soil	4056	7316	11372
	Total	23743	12585	36328
MGRVI	Vegetation	19625	2900	22525
	Bare soil	4118	9685	13803
	Total	23743	12585	36328
RGBVI	Vegetation	17410	3282	20695
	Bare soil	6333	9300	15633
	Total	23743	12585	36328
HSLVI	Vegetation	18634	8	18642
	Bare soil	5109	12577	17686
	Total	23743	12585	36328
HSLGEVI	Vegetation	21973	487	22460
	Bare soil	1770	12098	13868
	Total	23743	12585	36328

color space vegetation index, significantly reducing misclassifications and omissions. In summary, this study provides an effective monitoring mechanism and scientific decision support for desertification control in Hangjin Banner and similar areas through the strong synergy of satellite and RAV remote sensing.

APPENDIX

See Tables X and XI.

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